TECHNICAL REPORT · EXECUTIVE SUMMARY

DeepSeek-V3: Constraint-Driven Engineering at Scale

How 671B parameters were trained for $5.6M — and why the architecture matters beyond the headline.

CLIFFORD
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v1.0 · Internal

671B Total params
37B active

$5.6M Training cost
vs. $100M+ rivals

2,048 H800 GPUs
<2 months, zero rollbacks

DeepSeek-V3 is a proof that constraint drives innovation. Trained on export-controlled H800 GPUs with reduced interconnect, the team had no option to brute-force scale. Four architectural breakthroughs followed — none of which Western labs with unlimited H100 clusters ever bothered to discover.

Attention & Memory

Multi-Head Latent Attention (MLA) replaces the standard KV-cache with a compressed low-rank latent vector. Only this vector plus a small decoupled RoPE component are cached — dramatically reducing inference memory without performance loss.

- Full K/V pairs per head are not stored — latent compression handles it.
- Equivalent performance to standard Multi-Head Attention on all benchmarks.
- Enables long-context inference at 128K tokens without memory explosion.

MoE Routing

Auxiliary-Loss-Free Load Balancing solves the classic MoE problem: auxiliary losses corrupt the training signal. Bias terms added to routing scores are adjusted per-step to balance load, while the computation gating stays unbiased.

- 1 shared expert + 256 routed experts, 8 active per token.
- Bias-only routing: underloaded experts get boosted, overloaded get penalized.
- Ablations confirm superior quality vs. auxiliary-loss approaches.

Infrastructure & Precision

HPC co-design at the hardware level

FP8 TRAINING

First successful 671B-scale FP8

Tile-wise + block-wise quantization, CUDA Core promotion every 128 elements. Built around H800's limited 14-bit FP8 accumulation.

DUALPIPE

Near-zero comm overhead

Forward/backward interleaved from both pipeline ends. GPU SMs manually partitioned between compute and communication. Custom PTX for cross-node kernels.

MTP

Multi-Token Prediction

Predicts n+1, n+2, n+3 tokens per position, densifying training signal. Inference: discard for zero overhead, or use for speculative decoding at 1.8x TPS.

> **The real lesson is not the model — it is the methodology.** DeepSeek's $5.6M cost is evidence that the field has left massive efficiency gains on the table because brute force was always cheaper than ingenuity. When hardware access is constrained, optimization becomes architecture.

Sources: arXiv:2412.19437, DeepSeek-AI Technical Report, Dec 2024